

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XVII: THE TAIL NORM AT COMPOSITE q , AND THE SPLITTING OF THE LIMIT K -MODULE

RUQING CHEN

ABSTRACT. Paper VIII proved that for $q = p$ prime the kernel of the tail norm on the Iwahori tower is the full p -torsion of the sandpile group, and deduced from the resulting torsion-freeness that the sequence $0 \rightarrow \mathfrak{X}_\infty^\vee \rightarrow K_0(\mathcal{O}_\infty) \rightarrow \mathbb{Z}[1/q] \rightarrow 0$ splits. Both statements were left open for composite q , on the ground that $\mathbb{Z}/q\mathbb{Z}$ has zero divisors. That ground is illusory. The only ingredient of the prime proof that used primality is the rank identity $\text{rank}_{\mathbb{F}_p} A_{k+1} = n_k$, and A_{k+1} is the adjacency matrix of a line digraph, hence factors as $H^T T$ through the head and tail incidence matrices; the rows of H and of T are non-empty and disjointly supported, so the rank is n_k *in every characteristic*. With that, the two facts already known for all q — $|\ker \tau_*| = q^{n_k(q-1)-1}$ and $\ker \tau_* \subseteq \text{Jac}(Y_{k+1})[q]$ — close the problem by counting: the two orders are forced to agree, so

$$\ker(\tau_*: \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)) = \text{Jac}(Y_{k+1})[q] \cong (\mathbb{Z}/q)^{n_k(q-1)-1}$$

for every $q \geq 2$. Equality in the counting has a second consequence, which is new even for prime powers and is the sharpest statement here: *every invariant factor of $\text{Jac}(Y_{k+1})$ divisible by a prime $p \mid q$ is divisible by $p^{v_p(q)}$* . So at $q = 4$ the group has no element of order exactly 2, and the kernel of the tail norm strictly contains the 2-torsion — a structural phenomenon with no analogue in the prime case. Duality then gives the splitting for every q in three lines and without identifying $\mathfrak{X}_\infty$: $\mathfrak{X}_\infty[q] = 0$, so $\mathfrak{X}_\infty^\vee$ is q -divisible, so $\text{Ext}^1(\mathbb{Z}[1/q], \mathfrak{X}_\infty^\vee) = \varprojlim^1(\mathfrak{X}_\infty^\vee, \times q) = 0$. Six checks on eight towers ($q = 2, 3, 4, 5, 6$, complete and incomplete bases, to level three), all exact; the $q = 4$ towers could have falsified the divisibility statement and did not.

1. THE RANK IS CHARACTERISTIC-FREE

Throughout, Y_0 is a $(q+1)$ -regular graph on n vertices, Y_1 its non-backtracking directed-edge graph — Eulerian q -regular on $n_1 = n(q+1)$ vertices — and $Y_{k+1} = L(Y_k)$ the line digraph, so $n_{k+1} = qn_k$, for $k \geq 1$. The degeneracy maps τ (tail) and η (head) send the vertex $i \rightarrow j$ of Y_{k+1} to i and to j respectively; τ_* is the pushforward and η^* the pullback. We write

$$r_k := n_k(q-1) - 1,$$

$\Delta_k = qI - A_k$ for the out-Laplacian of Y_k , and $\text{Jac}(Y_k)$ for the torsion of $\text{coker } \Delta_k^\top$, so that $\text{coker } \Delta_k^\top \cong \mathbb{Z} \oplus \text{Jac}(Y_k)$. Three facts from the earlier papers hold for *every* $q \geq 2$ and will be used as stated:

- (i) $\tau_* \eta^* = A_k$, $\eta^* \tau_* = A_{k+1}$ and $\tau_* \tau^* = \eta_* \eta^* = qI$ [V, Prop. 1.2];
- (ii) τ_* is surjective on Jacobians with $|\ker \tau_*| = q^{r_k}$ [V, Thm. 3.1(ii)], a restatement of the line-digraph recursion $\kappa(Y_{k+1}) = q^{r_k} \kappa(Y_k)$ [III, Thm. 2.1];
- (iii) A_{k+1} acts on $\text{Jac}(Y_{k+1})$ as multiplication by q , and consequently $\ker \tau_* \subseteq \text{Jac}(Y_{k+1})[q]$ [VIII, Prop. 2.7].

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Only (iii) needs a word: $\Delta_{k+1} = qI - A_{k+1}$ annihilates the cokernel, so $A_{k+1} = q$ there, and by (i) $\tau_*x = 0$ forces $A_{k+1}x = \eta^*\tau_*x = 0$, i.e. $qx = 0$.

What was available only for $q = p$ prime is the rank of A_{k+1} modulo p [VIII, Thm. 2.8]. It is in fact insensitive to the characteristic.

Lemma 1.1 (Rank of a line-digraph adjacency matrix). *Let G be a digraph on n vertices and m arcs in which every vertex is the head of at least one arc and the tail of at least one arc, and let $A_{L(G)}$ be the adjacency matrix of its line digraph. Then $\text{rank}_F A_{L(G)} = n$ for every field F .*

Proof. Let $H, T \in F^{n \times m}$ be the head and tail incidence matrices, $H_{v,e} = [\text{head}(e) = v]$ and $T_{v,f} = [\text{tail}(f) = v]$. For arcs e, f ,

$$(H^T T)_{e,f} = \sum_v [\text{head}(e) = v][\text{tail}(f) = v] = [\text{head}(e) = \text{tail}(f)] = (A_{L(G)})_{e,f},$$

so $A_{L(G)} = H^T T$. Every arc has exactly one head, so the rows of H have pairwise disjoint supports; each is non-zero by hypothesis; hence they are linearly independent over any field and $\text{rank } H = n$, and likewise $\text{rank } T = n$. Thus $T: F^m \rightarrow F^n$ is surjective and $H^T: F^n \rightarrow F^m$ injective, so $\text{rank}(H^T T) = \dim H^T(F^n) = n$. $\square$

Corollary 1.2 (The p -rank of the Jacobian). *For $k \geq 1$ and every prime $p \mid q$, the p -rank of $\text{Jac}(Y_{k+1})$ equals r_k .*

Proof. $Y_{k+1} = L(Y_k)$ is q -regular, so $\Delta_{k+1} \equiv -A_{k+1} \pmod{p}$ and $\text{rank}_{\mathbb{F}_p} \Delta_{k+1} = n_k$ by Lemma 1.1. Tensoring coker $\Delta_{k+1}^T \cong \mathbb{Z} \oplus \text{Jac}(Y_{k+1})$ with $\mathbb{F}_p$ gives

$$n_{k+1} - \text{rank}_{\mathbb{F}_p} \Delta_{k+1} = 1 + \dim_{\mathbb{F}_p} (\text{Jac}(Y_{k+1}) \otimes \mathbb{F}_p),$$

and the left side is $qn_k - n_k$. As $\text{Jac}(Y_{k+1}) \otimes \mathbb{F}_p \cong \text{Jac}(Y_{k+1})[p]$ for a finite abelian group, the p -rank is $n_k(q-1) - 1 = r_k$. $\square$

2. THE TAIL NORM AT COMPOSITE q

Theorem 2.1 (The kernel of the tail norm, for every q). *Let $q \geq 2$ be arbitrary and $k \geq 1$. Then*

$$\ker(\tau_*: \text{Jac}(Y_{k+1}) \rightarrow \text{Jac}(Y_k)) = \text{Jac}(Y_{k+1})[q] \cong (\mathbb{Z}/q)^{r_k},$$

and moreover every invariant factor of $\text{Jac}(Y_{k+1})$ divisible by a prime $p \mid q$ is divisible by $p^{v_p(q)}$.

Proof. Write $e_p = v_p(q)$ and let the p -primary part of $\text{Jac}(Y_{k+1})$ be $\bigoplus_{j=1}^{s_p} \mathbb{Z}/p^{a_{p,j}}$ with $a_{p,j} \geq 1$; by Corollary 1.2, $s_p = r_k$ for every $p \mid q$. By (ii) and (iii) of Section 1,

$$q^{r_k} = |\ker \tau_*| \leq |\text{Jac}(Y_{k+1})[q]| = \prod_{p \mid q} p^{\sum_j \min(e_p, a_{p,j})} \leq \prod_{p \mid q} p^{e_p r_k} = q^{r_k}.$$

Both inequalities are therefore equalities. The first forces $\ker \tau_* = \text{Jac}(Y_{k+1})[q]$, two finite groups of the same order one contained in the other. The second forces $\min(e_p, a_{p,j}) = e_p$, that is $a_{p,j} \geq e_p$, for every $p \mid q$ and every j ; the p -primary part of $\text{Jac}(Y_{k+1})[q]$ is then $(\mathbb{Z}/p^{e_p})^{r_k}$, and taking the product over $p \mid q$ gives $(\mathbb{Z}/q)^{r_k}$. $\square$

Remark 2.2 (What was and was not primality). For $q = p$ this is [VIII, Thm. 2.8]. The present proof differs from that one only in Corollary 1.2, and the corollary differs only in proving the rank identity by a factorisation valid in every characteristic rather than by an argument in characteristic p . Nothing else in the chain ever meets a zero divisor of $\mathbb{Z}/q\mathbb{Z}$: the counting is carried out one prime at a time, and q enters only through its valuations.

Remark 2.3 (Prime powers behave differently from primes). For q squarefree, $\text{Jac}(Y_{k+1})[q] = \bigoplus_{p|q} \text{Jac}(Y_{k+1})[p]$ and Theorem 2.1 says only that the kernel of the tail norm is the torsion killed by each prime factor — the prime picture, assembled by the Chinese remainder theorem. For q not squarefree the theorem says something with no prime analogue. At $q = 4$ it asserts that $\text{Jac}(Y_{k+1})$ has *no element of order exactly 2*: every invariant factor divisible by 2 is divisible by 4. Consequently $\text{Jac}(Y_{k+1})[2] \subsetneq \text{Jac}(Y_{k+1})[4] = \ker \tau_*$, of index 2^{r_k} , and the kernel of the tail norm strictly contains the p -torsion for each $p \mid q$. This is a falsifiable statement about a concrete integer matrix; check (E) tests it on three $q = 4$ towers, including level three of the K_6 tower, where $\text{Jac}(Y_3)$ has 2-rank 359 and the smallest 2-adic valuation among its invariant factors is 2.

Corollary 2.4 (No q -torsion in the pro-module). $\mathfrak{X}_\infty := \varprojlim (\text{Jac}(Y_k), \tau_*)$ satisfies $\mathfrak{X}_\infty[q] = 0$.

Proof. Let $x = (x_k)_{k \geq 1} \in \mathfrak{X}_\infty$ with $qx = 0$. Then $x_k \in \text{Jac}(Y_k)[q]$ for every k , and for $k \geq 2$ that group is $\ker(\tau_*: \text{Jac}(Y_k) \rightarrow \text{Jac}(Y_{k-1}))$ by Theorem 2.1; hence $x_{k-1} = \tau_* x_k = 0$ for every $k \geq 2$, i.e. $x = 0$. $\square$

3. THE SPLITTING, FOR EVERY q

Theorem 3.1 (Splitting of the limit K -module). *For every $q \geq 2$ the short exact sequence of [VIII, Thm. 3.3(iii)],*

$$0 \longrightarrow \mathfrak{X}_\infty^\vee \longrightarrow K_0(\mathcal{O}_\infty) \longrightarrow \mathbb{Z}[1/q] \longrightarrow 0,$$

splits, and $K_0(\mathcal{O}_\infty) \cong \mathbb{Z}[1/q] \oplus \mathfrak{X}_\infty^\vee$.

Proof. The extension class lies in $\text{Ext}^1(\mathbb{Z}[1/q], \mathfrak{X}_\infty^\vee)$. Write $\mathbb{Z}[1/q] = \varinjlim (\mathbb{Z} \xrightarrow{q} \mathbb{Z} \xrightarrow{q} \cdots)$. For any abelian group B the Milnor sequence of this colimit reads

$$0 \rightarrow \varprojlim^1 \text{Hom}(\mathbb{Z}, B) \rightarrow \text{Ext}^1(\mathbb{Z}[1/q], B) \rightarrow \varprojlim \text{Ext}^1(\mathbb{Z}, B) \rightarrow 0,$$

and $\text{Ext}^1(\mathbb{Z}, B) = 0$, so $\text{Ext}^1(\mathbb{Z}[1/q], B) \cong \varprojlim^1 (B \xleftarrow{q} B \xleftarrow{q} \cdots)$. If B is q -divisible every transition map is surjective and this $\varprojlim^1$ vanishes. Now $\mathfrak{X}_\infty$ is profinite and multiplication by q on it is injective by Corollary 2.4; Pontryagin duality is exact, so multiplication by q on $\mathfrak{X}_\infty^\vee$ is surjective, i.e. $\mathfrak{X}_\infty^\vee$ is q -divisible. Hence $\text{Ext}^1(\mathbb{Z}[1/q], \mathfrak{X}_\infty^\vee) = 0$. $\square$

Remark 3.2 (A shorter route to the prime case). For $q = p$, [VIII, Thm. 3.3] obtained the splitting by first identifying $\mathfrak{X}_\infty^{(p)} \cong \mathbb{Z}_p^\mathbb{N}$ and then observing that its dual $(\mathbb{Q}_p/\mathbb{Z}_p)^{(\mathbb{N})}$ is divisible. The proof above needs neither identification: it uses only that multiplication by q is injective on $\mathfrak{X}_\infty$, which is Corollary 2.4. The structure of $\mathfrak{X}_\infty$ is not an input to the splitting, and for composite q we do not determine it.

4. VERIFICATION

The script `composite_tail_norm.py` runs the six checks of Table 1 in about five seconds; its output is archived as `composite_tail_norm_output.txt`. Eight towers are used: base graphs K_4 and Petersen ($q = 2$), K_5 and $K_{4,4}$ ($q = 3$), K_6 and $K_{5,5}$ ($q = 4$), K_7 ($q = 5$) and K_8 ($q = 6$), carried to level three where the size permits, the largest being Y_3 of the K_6 tower on 480 vertices. All arithmetic is exact: Bareiss determinants for the orders, and p -adic elimination (no coefficient explosion) for the valuations of the invariant factors.

Check (E) is the one that could have failed. For each $q = 4$ tower it reports the multiset of 2-adic valuations of the invariant factors of $\text{Jac}(Y_k)$: on K_6 at level two these are $2^{84}, 4, 5^4$ and at level three the 359 valuations are all at least 2; on $K_{5,5}$ at level two the 149 valuations are all at least 2. A single invariant factor of valuation 1 anywhere would have contradicted Theorem 2.1. Check (T) reproduces $\text{Jac}(Y_2) = (\mathbb{Z}/2)^8 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/16 \oplus \mathbb{Z}/48$ and $\text{Jac}(Y_3) = (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^8 \oplus \mathbb{Z}/8 \oplus \mathbb{Z}/32 \oplus \mathbb{Z}/96$ of [VIII, Table 1], with $\text{ord}_2 \kappa = 18$ and 41.

key	check	mode
(I)	the degeneracy identities of §1(i) hold verbatim at composite q	exact
(N)	$\kappa(Y_{k+1}) = q^{r_k} \kappa(Y_k)$, so $ \ker \tau_* = q^{r_k}$	exact
(R)	Lemma 1.1: $\text{rank } A_{k+1} = n_k$ over $\mathbb{F}_p$ for $p \mid q$ and $p \nmid q$ alike	exact
(P)	Corollary 1.2: the p -rank of $\text{Jac}(Y_{k+1})$ is r_k for $p \mid q$	exact
(E)	Theorem 2.1: every invariant factor divisible by p is divisible by $p^{v_p(q)}$	exact
(T)	the published prime- q values of [VIII] are recovered	exact

TABLE 1. The verification battery, 6/6.

5. STATUS, SCOPE, CORRECTIONS

Proved. Lemma 1.1, Corollary 1.2, Theorem 2.1, Corollary 2.4 and Theorem 3.1, for arbitrary $q \geq 2$ and $k \geq 1$. Together they close [VIII, Rem. 5.1(b)] in both of its parts.

Quoted, for every q and used as stated. The degeneracy calculus [V, Prop. 1.2]; the tail norm and its kernel order [V, Thm. 3.1], resting on the line-digraph recursion [III, Thm. 2.1]; the containment $\ker \tau_* \subseteq \text{Jac}[q]$ [VIII, Prop. 2.7]; the exact sequence [VIII, Thm. 3.3(iii)]; the Milnor $\varprojlim^1$ sequence and the exactness of Pontryagin duality, both textbook.

Remark 5.1 (Solved, open, impossible). *Solved:* the composite- q half of [VIII, Rem. 5.1(b)], in the strong form $\ker \tau_* \cong (\mathbb{Z}/q)^{r_k}$, together with the splitting for every q . *Open:* (a) the structure of $\mathfrak{X}_\infty$ itself for composite q — the splitting no longer needs it, and we have not determined whether $\mathfrak{X}_\infty^{(p)}$ is again a free $\mathbb{Z}_p$ -module when $p^2 \mid q$; (b) whether the divisibility statement of Theorem 2.1 — no invariant factor of valuation below $v_p(q)$ — has a direct combinatorial proof, independent of the counting, and what it says about the spanning-tree lattice of a line digraph; (c) the remaining items of [VIII, Rem. 5.1], which this note does not touch. *Impossible:* nothing new is shown impossible here. In particular the obstruction the task specification anticipated — that zero divisors in $\mathbb{Z}/q\mathbb{Z}$ would make the kernel smaller than $\text{Jac}[q]$ and force an order formula in terms of $\gcd(q, \cdot)$ — does not exist.

Remark 5.2 (Corrections to the task specification). Four, all in the same direction: the composite case is easier than expected, not harder. (1) The specification asked for the Smith normal form of A_{k+1} over $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$ and for a fallback order formula in terms of $\gcd(q, \text{Jac})$ should the kernel fail to be $\text{Jac}[q]$. Neither is needed: the kernel *is* $\text{Jac}[q]$, and the only rank computation required is over a field, where Lemma 1.1 makes it characteristic-free. (2) “Modulo q localisation has zero divisors” is true and irrelevant; no step localises at q . (3) The specification proposed expanding $\mathbb{Z}[1/q]$ as a direct limit of localisations over the primes $p \mid q$; the single tower $\mathbb{Z} \xrightarrow{q} \mathbb{Z} \xrightarrow{q} \cdots$ is what gives the $\varprojlim^1$ description, and splitting into primes is unnecessary. (4) The specification asked for the divisible subgroup structure of $\mathfrak{X}_\infty^\vee$. What is needed and true is only that $\mathfrak{X}_\infty^\vee$ is q -divisible, dual to $\mathfrak{X}_\infty[q] = 0$; no finer structure enters, and Remark 3.2 notes that this also shortens the published prime-case proof.

PROVENANCE

Unrefereed preprint. Every numbered statement is proved in full above from the three quoted facts of Section 1, each of which is stated in the earlier papers for all q ; the computations verify those inputs independently (checks (I) and (N)), the new lemma (check (R)), and the falsifiable consequence of the theorem (check (E)). The divisibility statement of Theorem 2.1 was found by the counting and then tested, not the other way round; the $q = 4$ towers were computed specifically because they could refute it.

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GUT GEOSERVICE INC., RIVIÈRE-BEAUDETTE, QUÉBEC, CANADA
Email address: `ruqing@hotmail.com`